12.9 Writing and Solving One-Step Addition and Subtraction Equations Within Real-World Contexts

Lesson Introduction

Benchmark	MA.6.AR.2.2 Write and solve one-step equations in one variable within a mathematical or real-world context using addition and subtraction, where all terms and solutions are integers.	Learning Target	• I can write and solve one-step equations in one-variable within a real-world context using addition and subtraction.

	Building On	Working Toward
Benchmark Coherence	MA.5.AR.2.4 Given a mathematical or real-world context, write an equation involving any of the four operations to determine the unknown whole number with the unknown in any position.	MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical or real-world context, where all terms are rational numbers.

Essential Questions	• How do you write an algebraic equation to represent a real-world situation? • How do you know what the variable represents in a real-world situation?
Lesson Narrative	Students will work to create one-step equations using real-world contexts. The lesson starts with having students look at the variety of ways an equation can be written and progresses to having students write their own equations using real-world scenarios.

Component Summary	12.9.1 Warm-Up (~5 min.)	Create an equation based on a word problem *(SE p. 327)*	12.9.4 Collaboration (10-15 min.)	Represent word problems with equations *(SE p. 330)*
	12.9.2 Collaboration (5-10 min.)	Write and solve equations in real-world contexts *(SE p. 327)*	12.9.5 Your Turn! (5-10 min.)	Write and solve algebraic equations representing real-world scenarios *(SE p. 331)*
	12.9.3 Guided Practice (10-15 min.)	Create algebraic equations based on word problems *(SE p. 328)*	12.9.6 Wrap-Up (~5 min.)	Reflect on confidence in the lesson's learning target *(SE p. 332)*

	Glossary	Materials	Additional Preparation
Resources	<u>Key Terms</u>: N/A <u>Additional Terms:</u> • Variable • Equation	N/A	N/A

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td colspan="2">Things to Consider</td></tr>
<tr><td>Students may not realize that equations such as $x + 4 = 9$ are equivalent to $9 - x = 4$ and $9 - 4 = x$.
Students may believe words like “less than” can only be characterized with subtraction (for example, see 12.9.2, question 1).</td><td colspan="2">Determine if 12.9.2 Collaboration and/or 12.9.3 Guided Practice would be better as whole class, small group, or partner activities for your students. If time is a problem, then consider skipping 12.9.4 Collaboration and/or assigning 12.9.5 Your Turn! for homework.
In the 12.9.4 Collaboration, students are creating their own word problem. This goes beyond the MA.6.AR.2.2 benchmark, but is doing so in an effort to help the student solidify their understanding of writing an equation from a context. Determine if this section will be helpful for your students or if this section needs to be scaffolded further for your students.
Students who can do the mathematics in their head may struggle with the purpose of writing an equation. These students may also write the equations as $x = 24 - 8$ because that is how they see the problem. For these students, it may be best to ask that they write two equivalent equations. Forcing the student to write $x + 8 = 24$ cannot be justified mathematically since their equation is equivalent. It may be helpful to have them state the expected form of the equations to be $x + p = q$ or $x - p = q$.</td></tr>
<tr><td colspan="2">Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td colspan="2">MLR 2: Collect and Display
Prepare students for discussion by asking them to first share their responses to “Why are your solutions correct?” with a partner. Listen for, collect, and display vocabulary and include any diagrams that students use to represent each equation. Throughout the lesson, continue to update collected student language and remind students to borrow language from the display as needed. This will help students use mathematical language during their paired and whole-group discussions.</td><td>MA.K12.MTR.7.1: Real-World Contexts
Students will be working to write and solve equations to represent real-world contexts. Clarifications within this standard emphasize connecting mathematical models to everyday experiences. Use this as an opportunity to provide students with practice determining if their equations are appropriate to solve the real-world problems described.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="3">Consider placing students in groups and assigning groups to each number of the 12.9.4 Collaboration. By differentiating which groups engage with each portion of the activity, the collaboration can become a jigsaw in which students are responsible for sharing their learning with others. When students are positioned as experts in their respective field (or question), they internalize more agency and accountability in what they are presenting.</td></tr>
<tr><td colspan="4">Lesson Notes:</td></tr>
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12.9.1 Warm-Up: Bell Work

Component Overview: *Students write an equation to represent a real-world situation. Writing an equation draws on previous lessons in which students wrote expressions to represent situations.*

12.9.1 Warm-Up: Bell Work

1. Before last night, Faith had p Instagram posts on her account. She made 4 more posts last night. She now has 313 Instagram posts on her account. Write an equation that can be used to find p, the number of posts Faith had on her account before last night.

   $p + 4 = 313$

Students may write an expression such as $313 - 4$ or the equation $x = 313 - 4$. Students often struggle writing equations for contexts where they can easily determine the solution. Help students understand that they are actually using algebraic reasoning because they understand what the unknown is. Allow students to share whatever they wrote without considering what is right or wrong. Use this information to inform your teaching and support students as they learn to write equations.

12.9.2 Collaboration: Solving Equations in Real-World Contexts

Component Overview: *Students will be collaborating to select an equation to represent a situation, describe the unknown, and then solve the equation. All equations will have only one variable and use addition or subtraction. Students will draw on their previous experience using inverse operations to solve equations.*

12.9.2 Collaboration: Solving Equations in Real-World Contexts

Work with your partner to complete the following.

1. Kiran's cat weighs 5 pounds less than Clare's puppy. Clare's puppy weighs 14 pounds.
   a. Select all of the equations that could be used to find the weight of Kiran's cat.
      - ☐ $x + 5 = 14$
      - ☐ $x - 5 = 14$
      - ☐ $x = 14 + 5$
      - ☐ $x = 14 - 5$
      - ☐ $x - 14 = 5$
      - ☐ $x + 14 = 5$

   b. What does the variable, x, represent in this situation?
      The pounds that Kiran's cat weighs

   c. What is the weight of Kiran's cat?
      9 *pounds*

2. Kiegan collected 7 more items than his cousin Joel did for the annual toy drive. Joel collected 12 items.

   a. Select all of the equations that could be used to find the number of items Kiegan collected.
      - ☐ $t = 7 - 12$
      - ☐ $t - 7 = 12$
      - ☐ $t + 7 = 12$
      - ☐ $t = 12 + 7$
      - ☐ $t + 12 = 7$
      - ☐ $t - 12 = 7$

   b. What does the variable, t, represent in this situation?
      The number of items that Kiegan collected

Consider working with students to identify the key parts of the word problem. Have students highlight, circle, or underline the information that may be helpful when writing the equation.

"Kiran's cat weighs 5 pounds less than Clare's puppy."

12.9.2 Collaboration: Solving Equations in Real-World Contexts (continued)

c. How many items did Kiegan collect for the toy drive?

19 items

3. Discuss with your partner what you notice about the equations that represent the two situations. Summarize your discussion.

Answers will vary. I notice that the equations can be written as addition or subtraction equations.

Encourage students to examine the process of writing an equation by asking the following questions:

- *Why could three equations represent the situation?*
- *How do each of the equations represent what is occurring in the problem?*

Students may incorrectly say that the variable t represents Kiegan instead of understanding the t represents the number of items Kiegan collects.

Specifically, have students make connections between the details of the real-world problem and the equation they select. Encourage students to manipulate each of the equations to see that they are equivalent.

12.9.3 Guided Practice: Writing and Solving Real-World Equations to Find a Value

Component Overview: *Students use real-world examples to explore writing an algebraic equation to represent a relationship, identifying the variable in the equation, and then solving the equation to answer a real-world question. This may be completed as a whole-group activity or with students working together in pairs or small groups.*

12.9.3 Guided Practice: Writing and Solving Real-World Equations to Find a Value

1. Jamal, a student at McLaughlin Middle School in Polk County, is preparing for a trip to Disney World with his mom and dad. An annual adult ticket is $\$139$. An annual child's ticket is $\$5$ less than an annual adult ticket. Jamal's parents are going to spend a total of $\$412$, before tax, on tickets.

   a. Write an algebraic equation to represent the relationship between the cost of a child's ticket and the cost of an adult ticket in this situation.
   Answers will vary. $139 - 5 = x$; $x + 5 = 139$

   b. What does the variable in your equation represent?
   The cost of a child's ticket

   c. Solve the equation.
   $x = 134$

   d. What was the cost of Jamal's ticket?
   $\$134.00$

2. After doing some additional research, Jamal's mom also decided to purchase the meal bundle for their trip. The meal plan price is different for adults than children. It costs $\$110$ for his mom and dad. The total for all three meal plans was $\$136$. An equation can be written to find the unknown cost of Jamal's meal plan.

   a. Write an algebraic equation that can be used to find the cost of Jamal's meal plan.
   Answers will vary. $110 + x = 136$; $136 - 110 = x$

   b. What does the variable in your equation represent?
   The cost of Jamal's meal plan

Consider allowing students to work on the 12.9.3 Guided Practice independently if they already have a firm grasp on writing and solving equations representing real-world situations.

Guide students through the connection of equivalent inverse equations that could represent either of the situations in questions 1 and 2. Remind students that they are solving to identify the unknown within each word problem and that variables represent the unknown numbers. Also, work with students to identify keywords within the word problems.

12.9.3 Guided Practice: Writing and Solving Real-World Equations to Find a Value (continued)

c. Solve the equation.
$x = 26$

d. How much was Jamal's meal plan?
$26.00

3. Use the equation $49 + t = 107$ to complete the following.

a. Write a word problem that can be represented by the equation $49 + t = 107$.
Answers will vary. Melinda sold 49 tickets so far and needs to sell t more to reach her goal of selling 107 tickets.

b. What does the variable represent based on the word problem you created?
Answers will vary. The number of tickets Melinda needs to sell to reach her goal.

c. What is the value of the variable?
$t = 58$

d. Share **only** your word problem with your partner. Use your partner's word problem to fill in the table.
Answers will vary.

Partner's Word Problem	What Does the Variable Represent?	Solve the Equation	Partner's Initials
If Cassey has read 49 pages of a book that has 107 pages, how many pages are left to read?	*The number of pages left to read*	$t = 58$	

Form a small group of students who need additional support. As you work with this group, consider reading the problem aloud to students and helping them identify and highlight keywords and numbers in the problems. Assist students as they use the keywords and numbers to construct and solve the equations. Encourage students to create and use visual aids, diagrams, and/or manipulatives as needed.

Scaffold the process of writing and solving a word problem for students as needed:

- *Select a real-life situation to write about.*
- *What does the 49 represent? What about 107?*
- *What does t represent?*
- *What inverse operation would you use to solve the equation?*
- *What does t equal?*
- *What does this represent in the word problem?*

12.9.4 Collaboration: Writing and Solving Equations for Real-World Contexts

Component Overview: *Students collaborate to write equations to represent word problems and then use the equations to solve for an unknown in each problem. Students may work with a partner on each problem, or each can complete alternating sections of the problems as they progress.*

12.9.4 Collaboration: Writing and Solving Equations for Real-World Contexts

Work with your partner to complete the following.

1. PJ and Jasmine were given the following word problem:

   Lexi's school is having a food drive. At the end of today's lunch period, Lexi collected a total of 34 non-perishable items for the food drive. Lexi celebrated, as it was 11 more than yesterday's entire lunch collection! How many total non-perishable items were collected yesterday?

Jasmine told PJ they needed to solve the problem using an addition equation. PJ disagreed and said they need to use a subtraction equation.

a. With your partner, determine who will follow Jasmine's suggestion and who will follow PJ's. Then, write the equation based on whose suggestion you are assigned and solve.
Answers will vary.

Jasmine's Suggestion "Addition"	PJ's Suggestion "Subtraction"
$x + 11 = 34$ $x = 23$	$34 - 11 = x$ $x = 23$

b. Compare your equations and discuss your solutions with your partner.
$23 + 11 = 34$
$34 = 34$

c. Explain who was correct in their initial suggestion – Jasmine, PJ, both of them, or neither of them.
Answers will vary. Both of them are correct because the equation can be written using addition or subtraction.

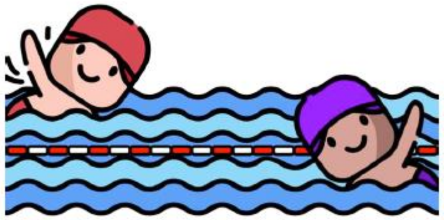
Consider assigning questions to groups of students based on readiness levels. Question 1 provides more scaffolding, as the word problem clearly lays out a scenario in which the equations will be exact. Question 2 may be more difficult for some students to access, as it requires creation of a scenario from which they must then also create an equation. Once the problems have been completed, engage the class in a jigsaw routine so that students can present their work to the class.

Throughout the 12.9.4 Collaboration, use the MLR 2: Collect and Display routine. Have students explain their reasoning for each question to their partner. Listen for, collect, and display vocabulary and include any diagrams that students use to represent each equation. Throughout Lesson 9, continue to update collected student language and remind students to borrow language from the display as needed.

12.9.4 Collaboration: Writing and Solving Equations for Real-World Contexts (continued)

d. How many non-perishable food items were collected during Lexi's lunch period yesterday?
23 *non-perishable food items*

2. An image of two swimmers in a pool is shown.



a. Work with your partner to create an addition or subtraction word problem based on the image.
Answers will vary. Swimmer 1 swam 3 more laps than Swimmer 2. If Swimmer 2 swam 9 laps, how many laps did Swimmer 1 swim?

b. Write an equation to represent your scenario.
Answers will vary. $3 + 9 = x$

c. What does the variable in your equation represent?
Answers will vary. The number of laps Swimmer 1 swam.

d. What is the value of the variable?
Answers will vary. $x = 12$

Encourage students to explain their reasoning by asking the following questions:

- *Why would you use an addition equation to represent the problem?*
- *Why would you use a subtraction equation to represent the problem?*
- *What does the variable represent in the equations?*

Challenge students to create a question in which a negative number must be used in the equation.

Encourage students to examine the equation by asking the following questions:

- *How does the equation relate to the word problem you wrote?*
- *How do the numbers in your equation represent what is described in your word problem?*
- *How can you use the equation to figure out the unknown?*

12.9.5 Your Turn!

Component Overview: *Students practice using addition and subtraction inversely in real-world scenarios. Students write an equation to represent each word problem and then use the equation to solve for the unknown.*

12.9.5 Your Turn!

Write and solve an algebraic equation for each of the following.

1. Priya and Gerri are working on an extra-credit poster for math class. Priya has 12 double-tip markers, which is 3 fewer than Gerri has. How many markers does Gerri have?

   $x - 3 = 12$
   $x = 15$

2. The most popular elective at Arts Academy is the improv class. Out of the 97 students that have applied, only 45 can be accepted. How many students are not accepted?

   $97 - 45 = x$
   $x = 52$

If students need support, consider allowing them to work with a partner. For students who need language support, read the word problems aloud to them. Encourage students to create diagrams or use manipulatives as needed.

12.9.6 Wrap-Up: Reflection

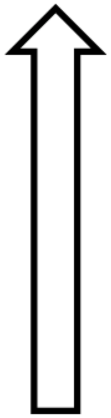
Component Overview: *Use this exercise as an opportunity to gather feedback on student progress from this lesson. This Wrap-Up assesses students' feelings about writing and solving one-step equations using addition or subtraction to solve real-world problems.*

12.9.6 Wrap-Up: Reflection

1. Shade in the arrow up to the statement that best describes your current understanding of the benchmark below:

   Answers will vary.

   Write and solve one-step equations in one variable within a mathematical or real-world context using addition and subtraction, where all terms and solutions are integers.



I'm so confident, I could explain how to write and solve one-step equations in one variable using addition or subtraction to someone else!

I can get to the right answer, but I don't understand well enough to explain how to write and solve one-step equations in one variable yet.

I understand some of this, but I don't understand all of it yet.

I have been trying hard, but I am finding this challenging.

Consider other resources for additional enrichment opportunities for students to practice or test their abilities.